Recording a Route Without Satellites

Speed from Vibration and Direction from Turns on a Smartphone, Measured Against GPS

Yu Yao-Hsing

euler.yu@gmail.com

Abstract

When GPS is lost — in a tunnel, an underground car park or an aircraft — a phone can still draw the route it travels. This paper describes a system that reads *speed* from how the vehicle shakes and *direction* from the phone's gyroscope, accelerometer and magnetometer, using GPS for nothing but the first point, and measures it against GPS on a motorcycle, in a car, on foot and on a plane. Every recording was replayed through the newest version from its raw sensor data, with GPS used only as the answer key: 116 motorcycle and car journeys (34 hours, 671 km that GPS could check), 129 straight stretches and 7 longer walks on foot, and three flights. Direction was within 30° of GPS 79% of the time on a motorcycle, 83% in a car and 93% on foot, and the median route was turned 9° on a motorcycle and 14° in a car. Against GPS, distance came out −9.9% on a motorcycle, −6.0% in a car and −0.7% on foot. In the air, where a smooth cabin says nothing about speed, the phone measures the takeoff roll and then takes the median speed of 1,496 real flights — NASA's public flight recorder data pooled with a day of OpenSky's worldwide ADS-B — at that many minutes since the roll and since the cabin started down, the cabin's pressure used only as a clock. On two four-hour flights graded by the aircraft's own ADS-B track it counted 95% and 94% of the distance, and on a short flight with GPS 78.4 of 77.4 km. Its direction starts from the takeoff roll's push matched to the departure runway and follows the gyroscope's turns, each checked against how far the magnetic field turned around the phone, because iOS quietly shortens a smooth turn on a phone lying still; over six recordings it was typically 13° off the track. A phone with no examples of its own starts with a small built-in network, which on recordings it had never seen was a little more accurate than a full store; both built-in networks are drawn layer by layer. We also report what each part of the direction method adds, where the system fails, and the approaches that did not work.

Keywords: dead reckoning; smartphone sensors; GNSS-denied navigation; vehicle speed estimation; heading estimation; magnetometer

1 Introduction

A phone that loses GPS still knows a great deal about how it moves. Its accelerometer feels every push and every bump, its gyroscope measures every turn, its magnetometer senses north, and its barometer feels height. In principle that is enough to keep drawing a route through a tunnel, an underground car park or a flight. In practice two things make it hard.

The first is *speed*. Adding up acceleration, the textbook answer, does not work on a phone: its small sensor errors grow into kilometres within minutes [1, 2]. The second is *direction*. The gyroscope measures turns but not north; the magnetometer measures north but in a vehicle it is disturbed by steel, and in any case it gives the direction the *phone* points, not the direction of travel, which differ by however the phone happens to sit in a pocket or a holder [3].

This paper describes a system built around those two problems, running as a mode of an ordinary iPhone app, and measures how well it works. Its contributions are:

- **A complete route recorder that uses GPS for the first point only.** Speed comes from a fingerprint of the vehicle's vibration, looked up in a store of the phone's own earlier trips or, on a new phone, answered by a small built-in network. Direction comes from the vehicle's turns, which reveal how the phone sits, on top of a heading whose slow drift is measured at stops and whose starting error is corrected by the magnetometer whenever the magnetic field proves to be Earth's, by holding still in the world while the vehicle turns (Section 3).
- **An evaluation against GPS that reflects the newest version, one kind of travel at a time.** Every recording was replayed through the current method from its raw sensor data (Section 3.5), and each kind of travel has its own section and figures: the motorcycle, walking, the car and the plane (Sections 4.1–4.4). Appendix A lists every journey.
- **How the built-in networks work.** The two small networks that answer before a phone has learned anything — one for speed on the ground, and a correction logged in the air — are drawn layer by layer, with how they were trained and how well they do (Section 5).
- **An account of what matters.** Every kind of travel side by side, what each part of the direction method adds (Section 6), where the system goes wrong, and the approaches that did not work (Section 7).

2 Related work

Inertial navigation. Integrating acceleration twice gives position, but the error grows quickly — with the square of time for an accelerometer bias and faster still for a gyroscope bias [1]; smartphone accelerometers and gyroscopes carry biases and noise large enough to make unaided integration useless within minutes [2]. Practical systems therefore constrain the problem.

Pedestrian dead reckoning. For walking, the usual constraint is the step: count steps, estimate their length and their direction [4]. A sensor on the foot can reset its velocity at every stance [5]; a phone in a pocket cannot, and its heading depends on how it is held, which several methods try to resolve [3]. Our walking mode follows the step-counting tradition; its direction method is our own.

Orientation. Orientation filters fuse gyroscope, accelerometer and magnetometer readings [6], and on an iPhone Apple's Core Motion provides one. Such filters assume gravity is the only steady force, so a vehicle's sustained acceleration — a takeoff roll — is mistaken for a tilt [7]. We found two more practical problems with the built-in heading, in a pocket and in a car (Section 3.2).

Vehicle speed from vibration. Road vibration changes with speed, and learned models can read speed from accelerometer data alone [8]. Our speed model is of that kind, but it learns continuously from the phone's own trips and must work for a phone in a trouser pocket on a motorcycle as well as in a car.

3 Method

A route is a series of short steps, one per second. Each step needs a length and a direction; with GPS both come from satellites, here both come from the phone (Figure 1).

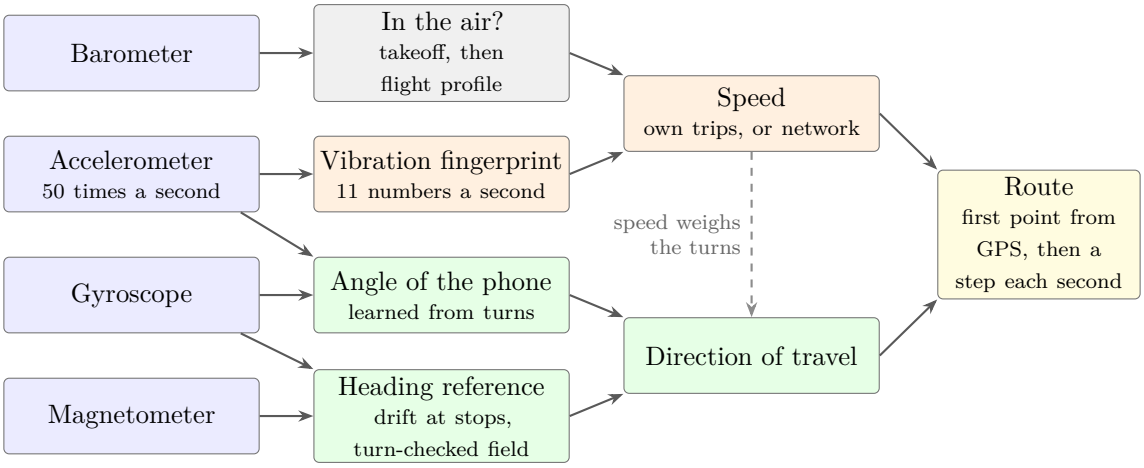


Figure 1. The system. Speed comes from the accelerometer's vibration, or in the air from integrating the takeoff and then from the typical speed of real airline flights; direction combines the angle of the phone, learned from the gyroscope's turns and the push felt in them, with a heading reference. GPS gives the first point and nothing else.

3.1 Speed

A fingerprint of the vibration. Every second the phone takes the last five seconds of up-and-down acceleration, measured 50 times a second, and reduces it to eleven numbers: the energy of the shaking in nine frequency ranges and two measures of its overall strength. A vehicle shakes differently at different speeds, and the fingerprint captures that.

A store of the phone's own trips. Whenever GPS is available the phone stores fingerprints with the speed GPS measured at that moment, keeping up to 4,000. Without GPS it looks up the 12 stored fingerprints closest to the current one and averages their speeds, nearer ones counting more:

$$\hat{v} = \frac{\sum_{j=1}^{12} v_j / d_j^2}{\sum_{j=1}^{12} 1/d_j^2}, \tag{1}$$

where v_j is a stored speed and d_j the distance between fingerprints. If even the closest stored fingerprint is far away, or the closest ones disagree among themselves, the store gives no answer rather than a guess, and the starting network below answers in its place: the speed never waits for GPS and never stalls while the vehicle moves. Averaging pulls every answer toward the middle, so a correction learned from the stored examples stretches the answers back out.

A starting network. A new phone has no examples, so until it holds 3,000 it uses a small network (11 inputs, two layers of 32, one output) trained on recorded trips and shipped with the app. It needs nothing from the current trip, and it declines a fingerprint unlike anything it was trained on.

Stops and hands. A stopped vehicle is recognised from the absence of shaking, not from the speed model, which would read an idling engine as a crawl. A hand on the phone shakes it, and shaking reads as speed, so while the phone is being handled the speed may fall but not rise, and a phone held in the hand keeps the last speed measured before it was picked up. That suits a car, where a passenger holds the phone at speed, but not a motorcycle, whose rider takes the phone out at a stop or on arrival. So when the trip so far looks like a phone in a rider's pocket — not held, turning at a median of at least 0.09 rad/s over at least a minute above 20 km/h, and not sitting upright facing its user — the held speed fades instead, to a third in 10 s.

In the air. A smooth cabin reads as standing still. In the air the phone instead measures the takeoff: from the last moment it was still, it follows its own turning with the gyroscope and adds up the push the accelerometer feels for up to two minutes. Core Motion's own orientation cannot be used for this, because it takes a takeoff's steady push for a tilt [7]. Only a takeoff starts the flight: the measured speed must pass 30 knots and then 200 km/h within 30 s, which the two flights did in 17 to 18 s and no road journey did in under 36. From that moment the measured speed is shown, and what was counted before it is made up.

After the takeoff: the typical flight. Integration beyond two minutes gathers too much error, and what a phone in a cabin can still sense without it is time: how long ago the takeoff roll began and whether the cabin has started down. The speed is then the median ground speed that real flights had at that point: in each minute since the roll, of the flights still climbing or cruising, until the median settles within 1% of the cruise (831 km/h, about 38 minutes in); and once the cabin has fallen 300 m below its highest level, in each minute since then. The difference the takeoff measured when it handed over is carried: most of it for two minutes and a little for twenty, as the flights themselves kept it (Section 4.4). The cabin's pressure is only a clock — airborne, the climb confirmed (400 m within ten minutes of leaving the ground), the descent, and the landing (back within 60 m of the ground, or 3% of its highest level, and no longer falling for 30 s; on both long flights 45 to 50 s after the touchdown) — and never a height: the cabin sits near 1.5 km while the aircraft cruises near 10. The height shown is the flights' typical height in the same way. Time is the real clock, so seconds in which iOS has paused the app are made up when it wakes. A flight whose takeoff the phone did not measure can be switched on by hand; the clock then starts from the cabin's height above the departure airport, read against the cabin's typical climb on the recorded flights (it rose almost identically on four phones, within 20 to 90 m for twelve minutes). No GPS, no tilt and no aircraft speed are used, and none of this runs on the ground.

Direction in the air. The compass is useless in a cabin: with the phone lying still, the field it measured turned by tens of degrees, and near a magnet it read many times Earth's. The direction starts instead from the takeoff roll, the strongest push a phone ever feels, straight along the runway: the average world-frame push over the roll's first 35 s. iOS measures that push from magnetic north, and on the return flight the same push read 194 to 215° on three devices on one runway, so it is matched to the runways of the airport at the phone's last GPS fix [11]: within 40° of one, the runway's direction is used. The gyroscope then follows the turns in episodes. A rotation counts when the cabin feels it — heavier than its own recent level, since an airliner banks to turn — or when it is slow and steady (0.12° a second for 15 s after a quiet minute): a 10° course change at cruise speed banks only 5°. An unfelt rotation lasting over two minutes is the gyroscope drifting; it is taken back and learned as the gyroscope's bias. A second in which the phone turns faster than 3.5° a second, or tilts by more than 5° that the gyroscope can account for, is a hand, and none of it is the plane's.

Core Motion's rotation rate is corrected for a bias that iOS learns whenever the device seems still, and a phone lying on an airliner's seat is still apart from a smooth turn of 1 to 2° a second. On the return flight the phone lying by a power socket read a median 64% of each turn the aircraft made, and then read the rest back the other way once the plane was straight; the phone in use read 98%. The magnetometer is not fooled: the horizontal field, in the phone's own frame, turned with the plane by 1.04 of each turn. Useless as a compass in a cabin, over the minute or two of a turn it measures how far the phone turned. So each turn is also followed in the field, and when the field stayed steady (its strength within 30%, its horizontal part 15 μT or more, the phone not handled) a felt turn takes the field's angle if the field turned 8° or more further the same way, or 10° or more the other way; a rotation the rules did not accept counts

when the gyroscope and the field both turned 8° the same way. Smaller field turns are left out: next to a magnet the field drifted 5 to 8° with no turn at all.

3.2 Direction

A heading reference. The phone's built-in heading follows a direction inside the phone that depends on how it is held, and when that direction points nearly straight down — a phone head-down at an angle in a trouser pocket — it scattered by about 49° a minute although the phone did not move. The system therefore computes its own heading from the phone's orientation, using whichever edge of the phone lies closest to level; in the same position that heading scattered by about 1° a minute.

Drift, measured at stops. In a car the heading also drifts slowly with the gyroscope's own bias, because the steel around it keeps the magnetometer from holding north: on one drive it turned 42° in 25 minutes with the phone fixed in place. A stopped car does not turn, so any turning the heading shows while the car is stopped and the phone still is drift. The phone measures it as a rate, at most 3° a minute, and takes the accumulated drift off every heading. It counts a still second only while the magnetic field reads like Earth's here, 30 to 60 μT: Core Motion's heading leans on the magnetometer, and in a disturbed field it is pulled toward the disturbance — which in a car is not the same standing as driving, so a drift measured at stops would be carried wrongly into the drive.

The magnetometer, when the field can be trusted. Core Motion also sets north once, in the first seconds of a recording, and then follows the gyroscope: on the recordings with the magnetometer logged its north moved by up to 160° in the first second and afterwards seldom more than 5° in a minute beyond what the gyroscope measured. Set inside a vehicle, that north is often wrong. On one drive it began 20–26° off while the magnetometer, read directly, put the phone within 1–5° of the truth. So the phone compares the magnetometer's heading with its own and takes the difference, less the drift already measured, as the correction to where the heading started — but only from a field that has shown it is Earth's. Earth's field is fixed in the world; a field from the car, or from something carried with the phone, turns with them. Over each last minute in which the heading swept at least 30°, the phone expresses the field in Core Motion's world frame; if it stayed within 8 μT of its mean, at a strength of 25 to 70 μT, that minute was Earth's field and its average difference is a sample of the correction. Until the first such minute, a field between 25 and 65 μT whose strength and dip have been steady for a minute is used instead. Beside a MagSafe charger the phone read up to 2,600 μT and nothing is taken from it.

The angle of the phone, from turns. The heading says where the phone points, not where the vehicle goes; the two differ by an angle β that depends on how the phone sits. In a turn the vehicle is pushed toward the inside of the curve, with a push equal to speed times turn rate, so comparing the sideways push the phone feels with the turning the gyroscope measures shows which way is forward. Written as complex numbers,

$$\hat{\beta} = \arg\left(\frac{\sum_R z_t}{\sum_R |v_t\omega_t|} + \frac{\sum_L z_t}{\sum_L |v_t\omega_t|}\right), \qquad z_t = \frac{a_t}{|a_t|}\,\overline{i\,v_t\,\omega_t}, \tag{2}$$

where a_t is the horizontal acceleration relative to the heading, written as its sideways and forward parts, v_t the speed, ω_t the turn rate, and R and L the seconds turning right and left. Braking into a corner and pulling away out of it lean the answer one way in right turns and the other way in left ones, so the two kinds are averaged apart. Only seconds above 14 km/h count,

Table 1. The recordings. GPS km is the distance GPS could check (†the four-hour flights' are their ADS-B tracks); half-minutes are the stretches used for speed; direction is graded in recordings with at least 50 graded seconds.

	Record-ings	Hours	GPS km	Half-minutes	Direction rec. (s)	Routes	Phone
Motorcycle	63	17.7	301	1372	65 (22,838)	66	trouser pocket
Car	53	16.4	370	1396	44 (22,627)	44	pocket, flat or mount
Walking	39	—	5.2	129	7 (721)	—	pocket
Plane	3	8.6	6,975†	—	6	5	hand, bag, seat

and only when the turn explains at least a tenth of the push; otherwise the push is braking or throttle, and a reversing car's push points backwards. The first estimate comes after 30 seconds of turning, and whenever the estimate improves the part of the ride already drawn is redrawn with it.

Walking. Over four seconds the back-and-forth of the steps lines up with the direction of walking, and each push-off is sharper than each landing, which tells which end of that line is forward. Steps are counted as peaks of the vertical acceleration and turned into distance with the stride Apple's pedometer has learned; the walk takes the larger of that and the pedometer's own total. In some pockets each footfall rings twice, with an echo about a quarter of a second after it, so the rhythm of the last four seconds is found from the signal's autocorrelation and, while it is clear, a peak sooner than 0.65 of its step period is refused as an echo.

Steps are also what ends a ride, and on a moving motorcycle the pedometer counts road bumps as steps: on one ride it counted 52 in 28 s at 41 to 56 km/h and handed the ride to walking. A pocket on a walking leg swings and one on a moving bike holds still, so while a vehicle is current, steps end it only once the phone's smoothed rotation reaches 0.60 rad/s; walking swings it at 0.8 to 3.9.

3.3 The route

After the first point, each second's step is laid from the previous point along the current direction, for the distance the speed covers in that second. No later GPS position, course or speed enters the route, the heading or the decisions between walking and riding; the magnetic declination is taken once, at the first point. Appendix B lists every setting (Table 11).

3.4 Recordings

All ground recordings were made by the author with one iPhone, mostly in one city, with the phone's raw sensor readings saved — acceleration, rotation, orientation and, in the most recent ones, the magnetic field, 50 times a second — and GPS recorded alongside. Table 1 summarises them. A motorcycle and a car were told apart from the sensors themselves: in a turn a motorcycle leans and the phone leans with it, while a car stays level and the phone is pushed sideways; this agreed with every journey whose vehicle was known. Walking is the time the app was counting steps. One car journey was recorded with the app measuring by GPS, so the app made no walking decisions in it; it was replayed as a car and, from the first step after the car stopped, as walking, as described when recorded, with GPS from every fix the phone logged.

Three flights are too few to learn the air speed from, so the typical flight is learned from two public sources. NASA's DASHlink archive of flight recorder data [9] holds 302 flights of

one regional jet in airline service with a takeoff, one value a second from the roll to landing. A day of the OpenSky Network's worldwide ADS-B state vectors [10], one report every 10 s, holds 1,426 takeoffs whose liftoff was received; 1,194 were by jets (1,034 single-aisle and 160 twin-aisle, by OpenSky's aircraft database), each followed through the day to its cruise and, for 1,079, down from the moment it was 300 m below its last cruise level, where the recorded cabin started down. OpenSky's flights begin at liftoff, so their clock starts NASA's median 25 s earlier and the roll's own minute is NASA's. Both are pooled, each flight counting once in each minute. The two four-hour flights, out and back on the same route in the same type of twin-aisle jet, had no GPS in the air; their answer is the aircraft's public ADS-B track, with the gaps over the sea filled by straight lines. Out, one phone was in a hand and one in a bag; back, one phone was in use, one lay by a power socket and a tablet lay on charge.

3.5 Measuring the newest version

So that every result describes the newest version, each recording was replayed through the current method from its raw sensor data:

- **Speed:** the same fingerprint, the same 12-neighbour average and correction, refusals answered by a starting network trained without that recording, and the same rules for stops and a phone in the hand, the fade on a pocket motorcycle included. The store holds 4,000 examples drawn at random from every *other* journey, so no journey is measured with its own GPS in the store. Each example keeps the same chance of being drawn whatever recordings are added, so a new recording changes the other journeys' stores only by its own examples. On 12 recent journeys the replay differed from what the app itself had drawn by a median 5%.
- **Direction:** the heading reference, drift correction, magnetometer correction (where the magnetic field was recorded), the angle learned from turns and the redrawing, with the replayed speed.
- **Starting network:** each recording measured by a network trained without it (the recordings split into eight groups), and the store tested holding only 60 to 4,000 examples.
- **Walking:** the step detector replayed on the recorded acceleration with the stride the phone had learned, and Apple's pedometer as logged where it was logged, modelled from the step count before that.
- **The flights:** the app's own code for the takeoff integration and test, the typical flight and the direction in the air, compiled and run on the flights' raw samples and the cabin's recorded pressure, against the short flight's GPS and the long flights' ADS-B tracks.
- **The typical flight, held out:** each NASA and OpenSky flight measured by medians built without its aircraft (the aircraft split into five groups), from the roll to landing.

Three inputs are taken from the recordings as they were: whether each second was walking or riding, Apple's motion classifier, and the car-park ramp flag. The exception is a ride that bump steps ended with the phone not swinging, which the newest version refuses: 37 such switches on 24 journeys, 23 minutes in all, are replayed as riding. Cabin pressure was not recorded on the short flight, so there the moment the app would call it airborne was taken from the same integration's climb rate.

In a pocket GPS is itself accurate only to 10–40 m, so it is used over stretches long enough for that not to matter.

- **Speed and distance** are compared over half-minute stretches between GPS fixes better than 40 m: the straight distance GPS moved against the distance the system added. For

walking, only straight stretches between fixes better than 25 m are used.

- **Direction** is compared every second: the bearing from the GPS position 15 s earlier to the one 15 s later, against the system's heading over the same half-minute (10 s for the plane).
- **Whole routes** are compared by finding the rotation and the size change that best lay the drawn route over the GPS track, pinned at the shared start, for routes of at least 1 km. The rotation is how far the route is turned; the size change how much too big or small it is; the drift is how far the drawn end lies from GPS's, as a share of the distance travelled.

10 car journeys went into, started in or ended in underground car parks. There GPS kept claiming 10 m accuracy for positions it did not have — frozen while parked, wandering on the ramps — so those stretches, found from the barometer, are left out of every comparison. So are half-minutes that mix walking and riding, and those in which the app counted steps while GPS showed riding speed, and the 47 seconds of one journey in which iOS had ended the app.

4 Results by kind of travel

Each kind of travel is shown on its own and measured the same way (Section 3.5): speed over half-minutes, distance over journeys, direction every second, and whole routes. Section 6 then sets them side by side.

4.1 Motorcycle

Figure 2 and Table 2 show the motorcycle, with the phone in a trouser pocket.

Table 2. Motorcycle at a glance (left), and speed by GPS speed band (right): the number of half-minutes, the middle GPS and app speeds, and the average error.

Recordings	
Journeys, hours	63, 17.7
Checked by GPS	301 km
Phone	trouser pocket
Speed and distance	
Half-minutes graded	1,372
Average speed error	8.8 km/h
Distance, all journeys	−9.9% (271 of 301 km)
Median journey	−6.3%
Journeys within 10%, 20%	50%, 73%
Range, app/GPS	0.33–1.29
Direction	
Graded	65 rec., 22,838 s
Median error	13°
75th, 90th percentile	27°, 49°
Within 15°, 30°, 45°	54%, 79%, 89%
Whole routes, at least 1 km	
Routes	66, 544 km
Turned, median	9°
Within 15°, 30°	70%, 88%
Turned more than 45°	4
Size, median	0.85 (61% within 20%)
End drift, median	14% of distance

GPS band		Middle (km/h)		Error
(km/h)	n	GPS	App	(km/h)
0–10	399	2.6	2.3	4.2
10–20	263	15.3	19.7	8.5
20–30	236	25.3	28.8	7.3
30–40	189	34.7	32.1	6.4
40–50	138	44.0	36.0	11.3
50–60	83	54.4	38.4	18.7
60–80	59	64.8	38.8	29.4
80+	5	98.4	40.9	67.7
All	1372	21.1	23.5	8.8

Speed. The app squeezes the range: 10–20 km/h reads about 20, 20–30 km/h about 29 where GPS measured 25, and above that it reads ever lower: about 32 at 30–40 km/h and about 39 where GPS measured 65 at 60–80 km/h. The squeeze comes from the averaging: when a fingerprint could belong to several speeds, the average lands in the middle. The average error is 8.8 km/h over 1,372 half-minutes.

Distance. The app added 271 km where GPS measured 301 km, 10% short: the distance lost at high speed outweighs what is gained at low speed. Journey by journey the median was 6.3% short, and 73% of journeys came within 20% of GPS.

Direction. Over 65 recordings the median error is 13° and 79% of seconds are within 30°. The tail is long: one second in ten is off by more than 49°, because a phone in a trouser pocket moves as the rider shifts. In the first minutes the angle of the phone is still being learned and the error is highest; it settles within about five minutes. After that the full method and the compass heading with the angle from turns alone are close, since a motorcycle's stops rarely hold the phone still enough to measure drift.

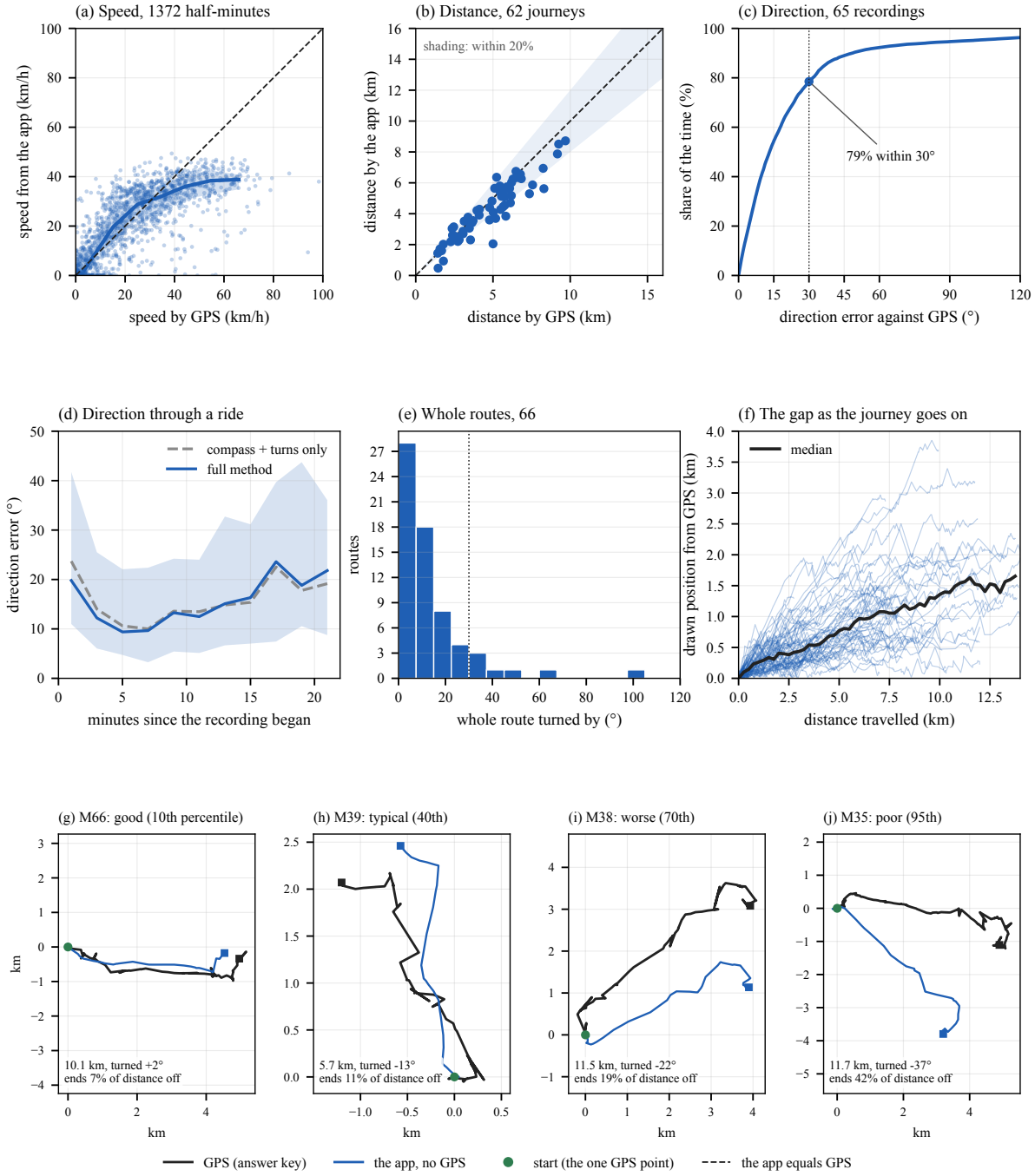


Figure 2. Motorcycle, with the phone in a trouser pocket. (a) Speed per half-minute against GPS; the line is the middle of each 10 km/h band, the shading the middle half, and on the dashed line the two agree. (b) Distance per journey of at least 1 km; the shading is within 20%. (c) How often the direction is within a given angle of GPS. (d) Direction error through a ride, in two-minute bins with at least 300 graded seconds: the full method against the compass heading with the angle from turns alone. (e) How far each whole route is turned. (f) How far the drawn position is from GPS as the journey goes on: one thin line per route, and the median. (g–j) Four routes at the 10th, 40th, 70th and 95th percentile of end drift, labelled as in Appendix A; each is turned together with its GPS track by an arbitrary angle so as not to show where it was recorded.

Whole routes. The median route is turned 9° and 88% are within 30°. Routes come out small — a median 0.85 of the real size — partly because distance is short and partly because small direction errors fold the path, and the drawn end lies a median 14% of the distance travelled from GPS's. Routes turned more than 45°: 4, of them 3 under 3 km, with too few turns to learn the angle of the phone well.

4.2 Walking

Walking is counted by steps, with the stride the phone has learned (Figure 3, Table 3). Over 129 straight half-minutes the app read about 4.8 km/h where GPS measured 4.5, and counted 1% less distance than GPS: 3% more in the first minutes after a ride and 8% less at other times. Direction comes from the back-and-forth of the steps: over 7 longer walks the median error is 10° and 93% of seconds are within 30°.

On 39 of the 129 stretches the steps counted ran more than 15% faster than the walking rhythm itself; on the walk after one motorcycle ride the count was 2.5 to 2.75 steps a second against a rhythm of 1.85, and those stretches came out close to twice what GPS saw. Refusing the echo removes most of the extra steps. The app takes the larger of its own count and Apple's pedometer, and the pedometer was not recorded before the check, so on the older walks the re-run models it from what each version logged and its walking figures there do not move with the check. Scaled by the change in the count alone, those 49 stretches would go from +9.1% to about +0.3% and the ones within 20% from 43% to 53%; 26 stretches come closer to GPS and 4 move away, all of them already short. The walks recorded since, with the check and the pedometer both live, need no estimate: the re-run reads their pedometer as logged, and as the app drew them on the day, over 70 stretches on 20 journeys, it counted 0.7% less distance than GPS, 59% of the stretches within 20%.

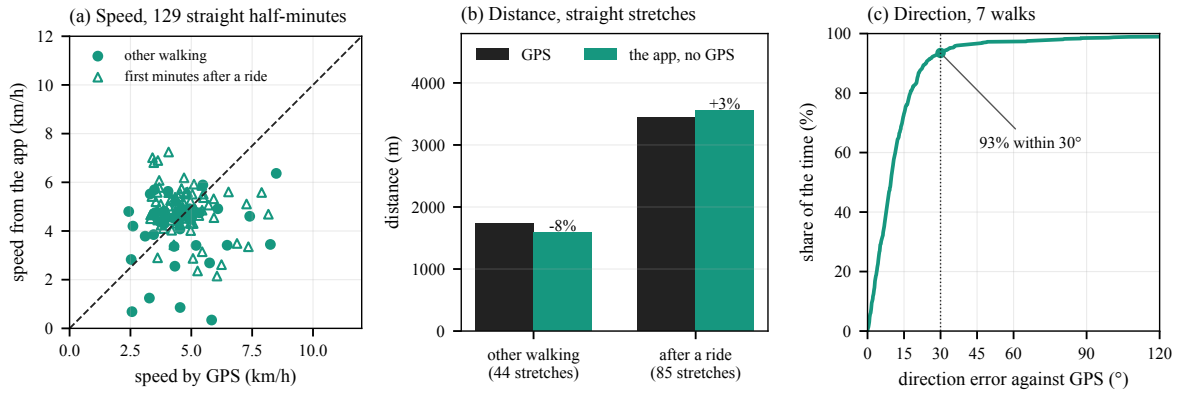


Figure 3. Walking, with the phone in a pocket. (a) Speed over straight half-minutes against GPS; triangles are the first minutes after a ride. (b) Distance over the straight stretches, added up. (c) How often the direction is within a given angle of GPS, over 7 longer walks.

Table 3. Walking: straight half-minute stretches between GPS fixes better than 25 m.

		Distance (m)			Speed (km/h)	
Straight stretches	n	GPS	App	Difference	GPS	App
Other walking	44	1740	1592	−8.5%	4.5	4.6
First minutes after a ride	85	3440	3553	+3.3%	4.6	4.9
All	129	5179	5144	−0.7%	4.5	4.8

4.3 Car

Figure 4 and Table 4 show the car, with the phone in a pocket, lying flat or in a mount.

Table 4. Car at a glance (left), and speed by GPS speed band (right): the number of half-minutes, the middle GPS and app speeds, and the average error.

Recordings	
Journeys, hours	53, 16.4
Checked by GPS	370 km
Phone	pocket, flat or mount
Speed and distance	
Half-minutes graded	1,396
Average speed error	7.8 km/h
Distance, all journeys	−6.0% (348 of 370 km)
Median journey	+1.1%
Journeys within 10%, 20%	53%, 77%
Range, app/GPS	0.57–1.57
Direction	
Graded	44 rec., 22,627 s
Median error	14°
75th, 90th percentile	24°, 56°
Within 15°, 30°, 45°	51%, 83%, 88%
Whole routes, at least 1 km	
Routes	44, 381 km
Turned, median	14°
Within 15°, 30°	57%, 86%
Turned more than 45°	1
Size, median	0.99 (86% within 20%)
End drift, median	17% of distance

GPS band		Middle (km/h)		Error
(km/h)	n	GPS	App	(km/h)
0–10	372	2.7	2.6	3.4
10–20	227	15.0	14.8	6.2
20–30	207	25.2	24.9	5.8
30–40	170	35.0	34.8	6.3
40–50	111	44.5	45.4	7.1
50–60	91	54.6	57.4	7.6
60–80	129	68.7	65.0	10.2
80+	89	93.4	62.0	35.7
All	1396	25.0	25.4	7.8

Speed. Much closer than on the motorcycle: from 10 to 60 km/h the middle reading is within about 3 km/h of GPS. Above that it reads low: 65 where GPS measured 69, and 62 where it measured 93. The average error is 7.8 km/h over 1,396 half-minutes.

Distance. The app added 348 km where GPS measured 370 km, 6.0% short. Journey by journey the median was 1.1% long, and 77% of journeys came within 20% of GPS.

Direction. Over 44 recordings the median error is 14° and 83% of seconds are within 30°; one second in ten is off by more than 56°. The full method stays below the compass heading with the angle from turns alone for most of a drive, as drift measured at stops and the magnetometer keep the heading reference in place.

Whole routes. The median route is turned 14° and 86% are within 30°. Car routes have the right size (median 0.99), and the drawn end lies a median 17% of the distance travelled from GPS's. Routes turned more than 45°: 1, of them 0 under 3 km.

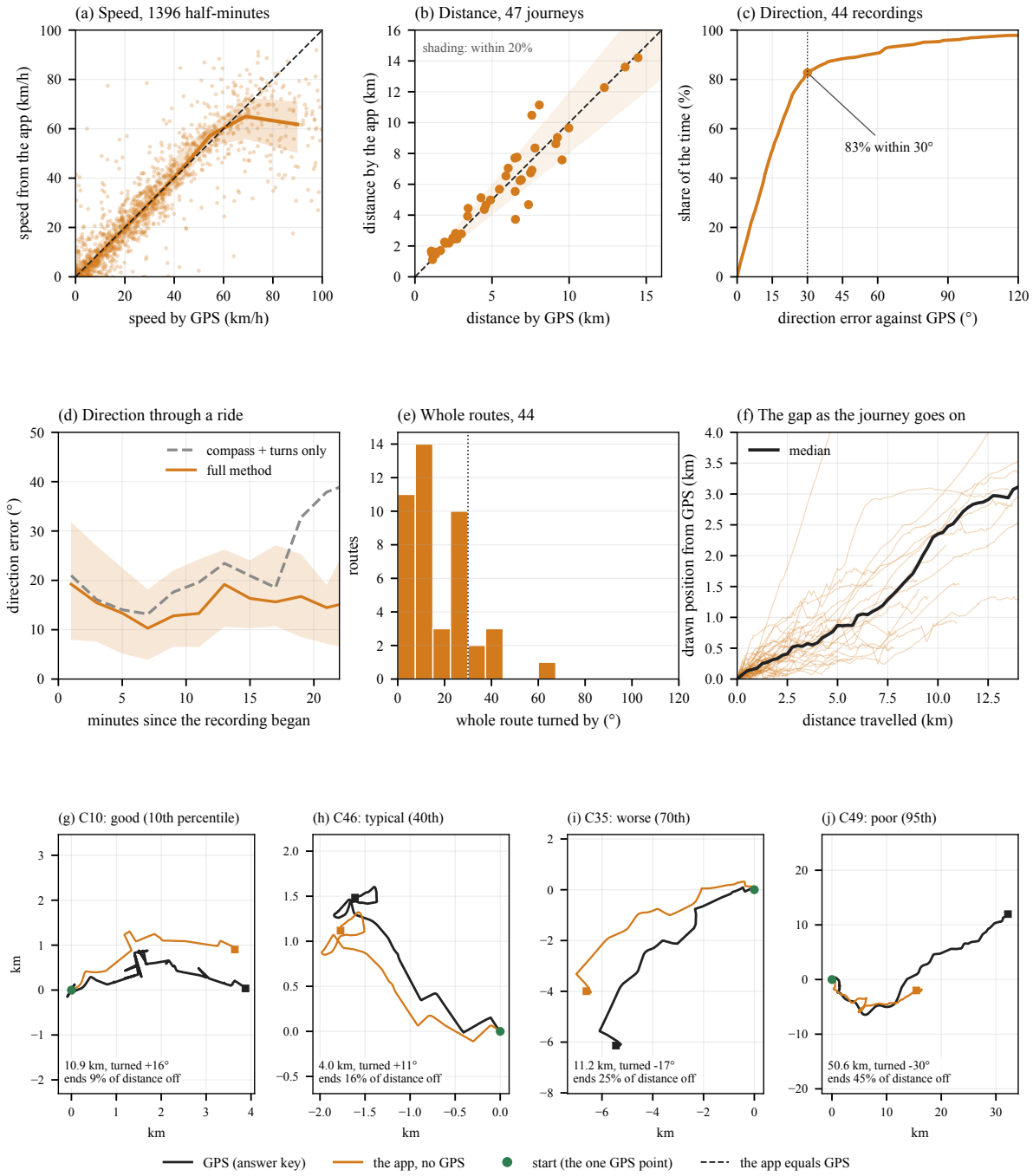


Figure 4. Car, with the phone in a pocket, lying flat or in a mount. (a) Speed per half-minute against GPS; the line is the middle of each 10 km/h band, the shading the middle half, and on the dashed line the two agree. (b) Distance per journey of at least 1 km; the shading is within 20%. (c) How often the direction is within a given angle of GPS. (d) Direction error through a ride, in two-minute bins with at least 300 graded seconds: the full method against the compass heading with the angle from turns alone. (e) How far each whole route is turned. (f) How far the drawn position is from GPS as the journey goes on: one thin line per route, and the median. (g–j) Four routes at the 10th, 40th, 70th and 95th percentile of end drift, labelled as in Appendix A; each is turned together with its GPS track by an arbitrary angle so as not to show where it was recorded.

4.4 Plane

On the ground the plane is measured like a car: while taxiing the app reads about 12 km/h where GPS measured 18. In the air a smooth cabin reads as standing still, so the phone measures the takeoff — 265 km/h forty seconds into the roll on the short flight, where GPS measured 266 — and then follows the typical flight (Figure 5, Table 5).

On the four-hour flight, graded by the aircraft's ADS-B track (3,392 km), the pooled flights counted 95% of the distance and NASA's jet alone 87%; the typical error was 40 km/h in the climb, 61 in the cruise and 29 in the descent, against 81, 132 and 58. The tilt network the app used before counted 80% on the phone in a hand and 39% on the one in a bag, which turned over at the takeoff push and read a tilt beyond anything NASA's flights had. On the short flight with GPS (77.4 km from the roll) the app counted 78.4 km, 17 km/h off GPS in minutes 1 to 3 and 15 in minutes 5 to 10; held at the measured speed, the flight counts 48.6 km. On the return flight (3,506 km by ADS-B), as the phone that measured its takeoff showed it with the landing as the cabin now reads it, the app counted 94%, typically 80 km/h off in the climb, 44 in the cruise and 102 in a descent that took fifty minutes, longer than the typical flight's. Held out by aircraft, the pooled flights came within 10% of the distance on 72% of OpenSky's jets and 36% of NASA's flights.

The speed is a typical airliner's, not this one's. The twin-aisle jets cruised near 905 km/h and the single-aisle ones near 820 (Figure 5c); the four-hour flight's twin-aisle jet flew 865 to 902 and the pooled cruise read 61 km/h low. A phone cannot tell which it is on. What it measures of its own aircraft is the takeoff, and that is carried: across the flights a difference measured 1.8 minutes after the roll was still 86% there half a minute later, 58% after a minute, about a third from two to four minutes and a few percent at twenty, the same in NASA's once-a-second data and in OpenSky's. Height, from the same flights, was typically 0.6 km off in the climb, 1.2 in the cruise and 1.2 in the descent; the cabin's pressure, read as a height, would have been 8.4 km off in the cruise. A network trained on the same pooled flights and the same two clocks came within 10% on 76% and 48% held out, ahead mostly in the descent, but read the short flight's first minutes 42 km/h off GPS; the app logs it beside the medians and uses the medians.

Over the six recordings the heading was typically 13° off the track, against 39° before, and within 30° of it 72% of the time; on the five phones 8° against 25° (Figure 6). Back, the phone in use was 6° off (17° before) and the one by the socket 5° (49°): its push read 12° off the runway, and the field restored the turns iOS had shortened, 94° of the 209° orbit before the landing among them. Out, the phone in the bag was 13° off (26°), and the short flight 5° (7°). Two did worse or stayed poor. The phone in the hand on the way out was 20° off against 9°: iOS paused the app for 29 and 25 minutes while the plane turned 35 and 8°, which no sensor reading afterwards recovers, and the earlier rule came out ahead only by counting the phone being handled as a 15° turn. The tablet, handled through the takeoff and lying in a field that a charger swung by hundreds of degrees, started the climb far off and stayed 55° off (138°).

Figure 7 puts speed and direction together. Drawn from the runway with nothing but what the phone senses, the return flight ended 217 km from the airport on the phone in use and 251 km on the one by the socket, against 889 and 2,453 km before; out, the phone in the hand ended 601 km off (214 before) and the one in the bag 419 km from where the plane was when it stopped (845); the short flight ended 6.3 km from where GPS ended, against 9.7 km. What remains is mostly the speed: the app reads a typical airliner, and a tailwind of 50 km/h over four hours is 200 km.

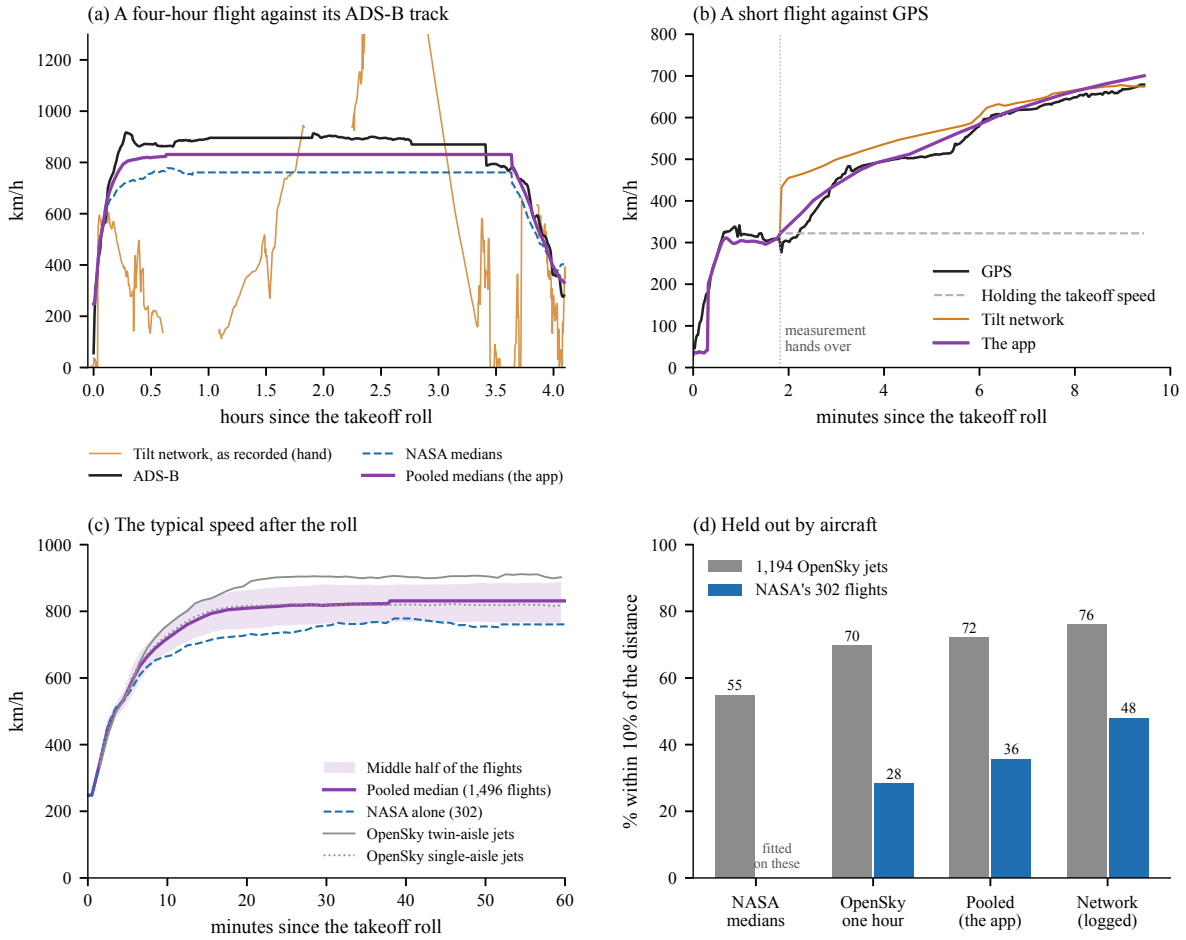


Figure 5. Speed in the air. (a) The four-hour flight against its ADS-B track: the typical flight of NASA's jet, the pooled flights the app uses, and the tilt network the app used before, as the phone in a hand recorded it. (b) The short flight against GPS: the takeoff is measured until the dotted line, then the typical flight carries what it measured; holding the measured speed misses the climb. (c) The typical speed after the roll: the pooled median and the middle half of the flights, NASA's jet alone, and OpenSky's twin- and single-aisle jets. (d) Flights within 10% of the distance from the roll to landing, each measured without its own aircraft.

Table 5. Speed in the air. The four-hour flight against its ADS-B track (errors are the typical km/h in the climb, the cruise and the descent), the short flight against GPS from the roll, and flights held out by aircraft. [a]As the phone in a hand recorded it, pauses by iOS included.

	Four-hour flight (ADS-B)		Short flight (GPS)			Held out, within 10%	
	Distance	Error	km of	Error, minutes		OpenSky	NASA
In the air		climb/cruise/desc.	77.4	1–3	5–10	jets	flights
Hold the takeoff speed	—	—	48.6	17	301	—	—
Tilt network, before	80%[a]	—	82.0	77	20	—	—
NASA medians	87%	81/132/58	77.3	24	9	55%	fitted
Pooled (the app)	95%	40/61/29	78.4	17	15	72%	36%
Network (logged)	95%	38/58/28	81.6	42	32	76%	48%

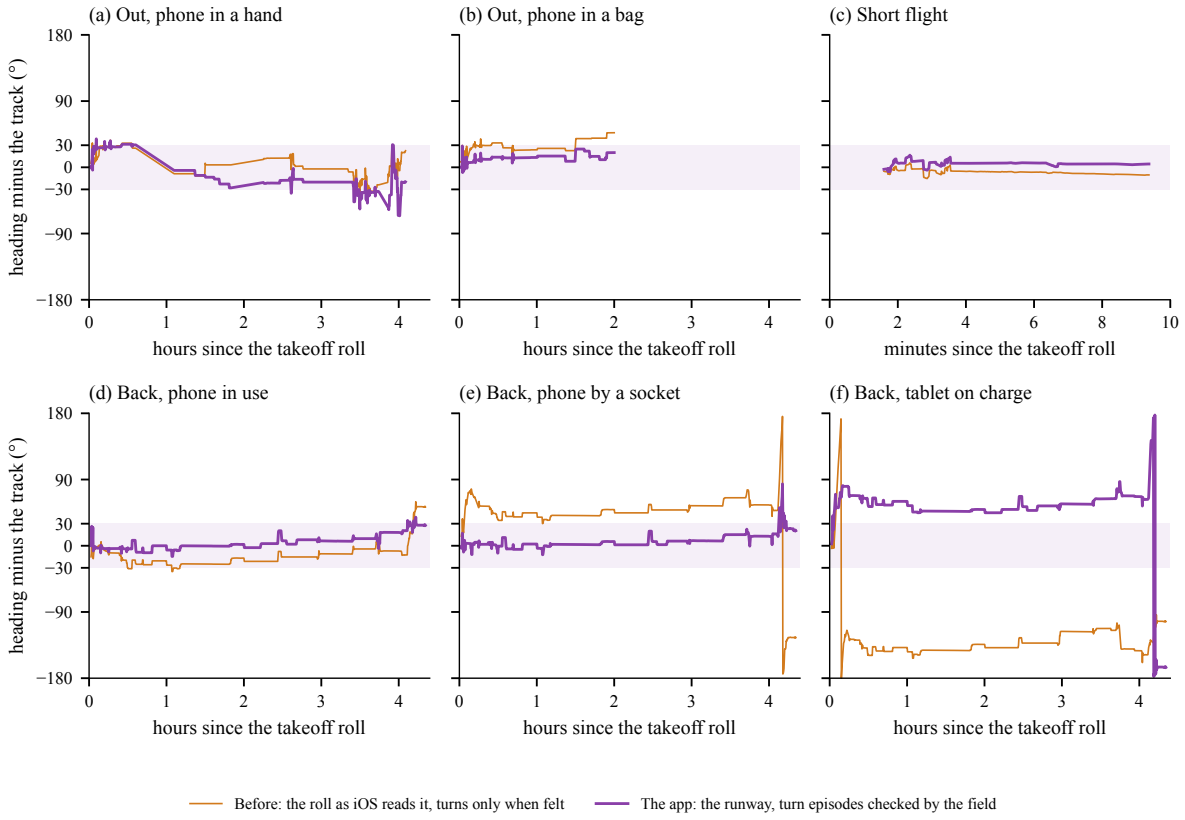


Figure 6. Direction in the air: the heading minus the aircraft's track (ADS-B; GPS on the short flight), as the app's own code computes it when replayed on each recording, before (the roll in iOS's frame, turns only when felt) and now (the roll matched to the runway, turn episodes checked against the field); the band is ±30°. Out: the phone in the bag was stopped after two hours. Back: one phone in use, one lying by a power socket, a tablet on charge that was handled through the takeoff.

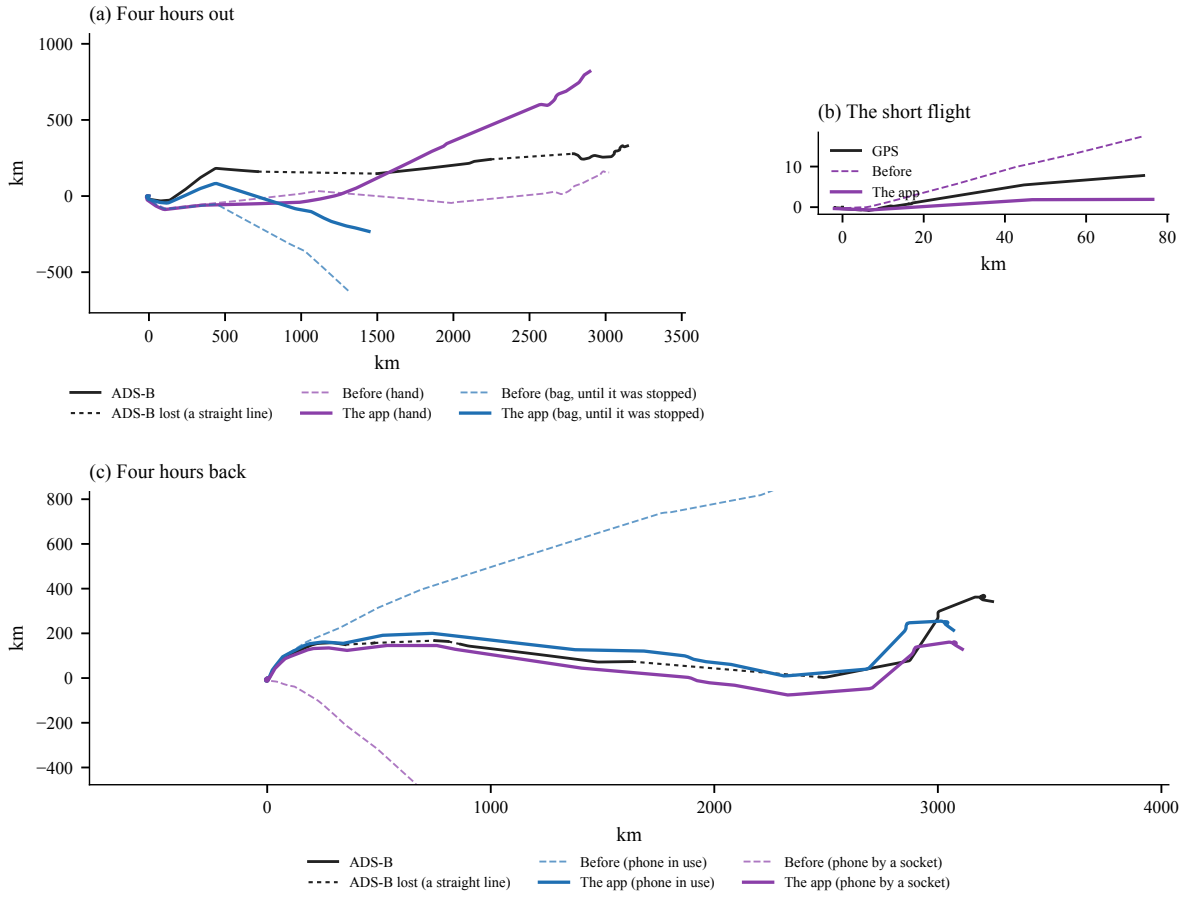


Figure 7. The flights drawn with no GPS after the first point: the app's speed, the same for every device of a flight, with each device's heading before (dashed) and now. (a) Out: ADS-B (dashed where the receivers lost it over the sea), the phone in a hand and the one in a bag until it was stopped. (b) The short flight against GPS. (c) Back: the phone in use and the one by the socket; the tablet is left out, its route far off the map. Like the example routes in Figures 2 and 4 they are turned by an arbitrary angle, on a plain km grid, so as not to show where they were recorded.

5 A new phone: the built-in networks

A phone that has learned nothing yet still has to answer. Two small networks ship with the app for that: one reads speed from the vibration fingerprint on the ground; the other, in the air, is a correction on top of the typical flight, which the app logs beside it but does not use. Figure 8 draws both.

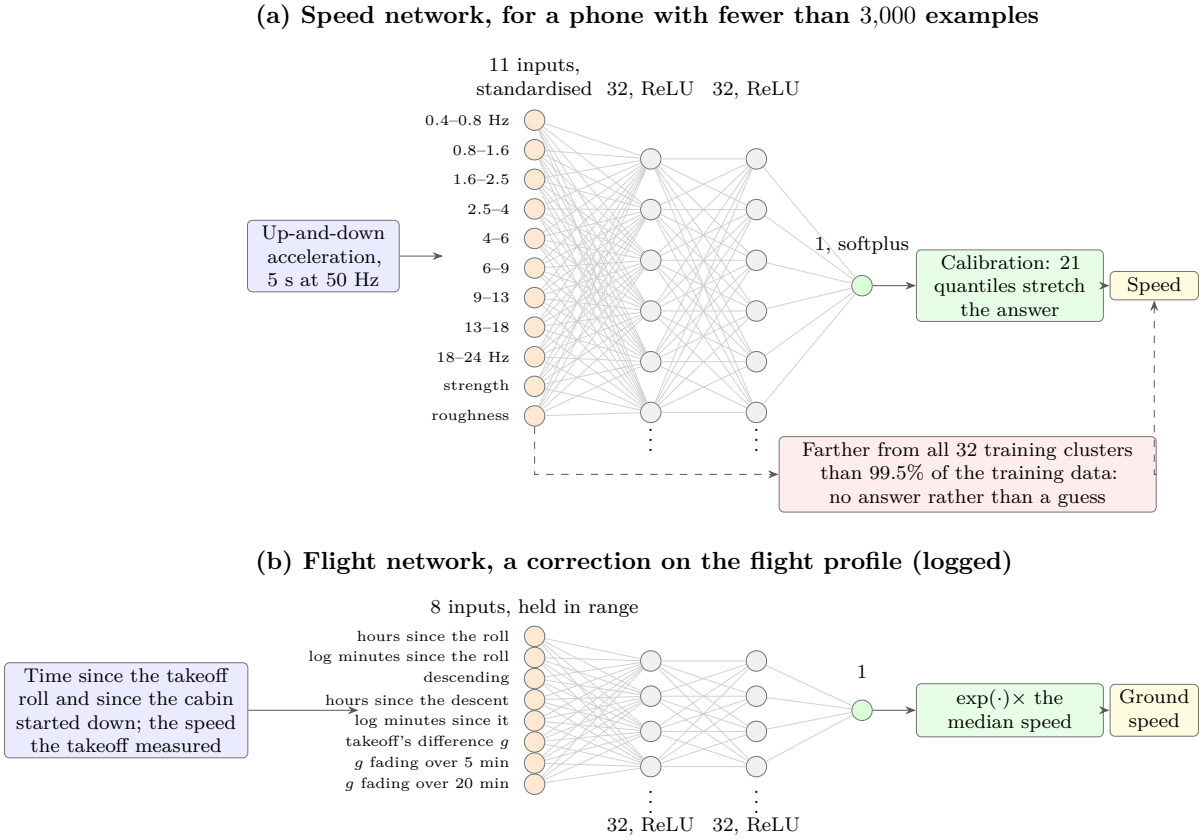


Figure 8. The two built-in networks, drawn layer by layer; each hidden layer has 32 units, of which only some are drawn. (a) The speed network: the fingerprint's eleven numbers go through two layers of 32 and one output, which a calibration then stretches; a fingerprint unlike any it was trained on gets no answer. (b) The flight network: the time since the takeoff roll and since the cabin started down, and the difference the takeoff measured, go through the same two layers; the output scales the typical speed.

5.1 How the speed network works

Its inputs are the fingerprint of Section 3.1: the energy of the up-and-down shaking in nine frequency bands from 0.4 to 24 Hz, and two measures of its overall strength — its spread, and how sharply it changes from one sample to the next. Each is standardised with a mean and scale fixed from the training data, never from the phone's own store, so clearing the store cannot change the network. Two layers of 32 units with ReLU activations follow, then one output unit with a softplus, which keeps the answer above zero: 1,473 numbers in all, evaluated once a second.

An averaging model pulls its answers toward the middle, so the output goes through a calibration, as the store's does: the network's answers on its training data at 21 quantiles are matched to the true speeds at the same quantiles, and each new answer is mapped along that line. And like the store it may refuse: the training fingerprints were grouped into 32 clusters, and a fingerprint farther from every cluster centre than 99.5% of the training fingerprints gets

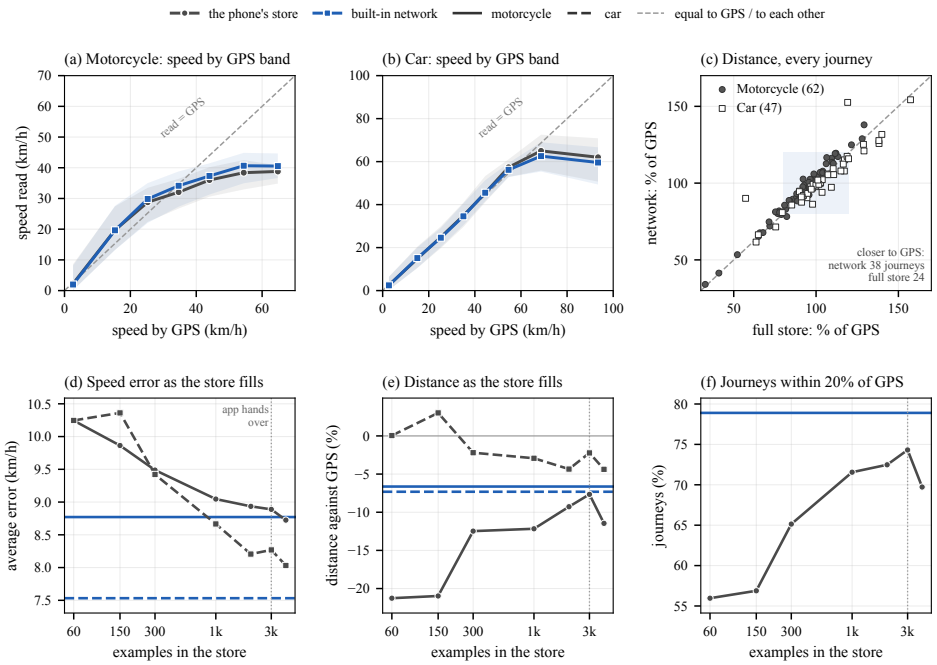


Figure 9. A new phone: the built-in network (blue, squares) against the phone's own store (grey, circles) on the same journeys, the network only on recordings it had never seen and the store never holding a journey's own examples. (a, b) The speed read in each 10 km/h band of GPS speed, the middle of the band and its middle half shaded, with a full store of 4,000 examples; on the dashed line the reading equals GPS. (c) Every journey of at least 1 km: the share of GPS's distance counted with the network against the share with the full store; on the dashed line the two agree, and in the shaded square both are within 20%. (d–f) A store still filling up, from 60 to 4,000 examples, against the network (horizontal lines): the speed error and the distance for each vehicle (solid: motorcycle, dashed: car), and the journeys within 20%; the dotted line is where the app hands over. In the air neither is used.

no answer rather than a guess.

It was trained on the GPS-labelled fingerprints of the recorded trips, with a Huber loss (which weighs large errors less than a squared loss does), the Adam optimiser and 40 passes over the data. The app uses it until the phone's own store holds 3,000 examples. For the results below each recording is measured by a network trained without it, the recordings split into eight groups.

5.2 How the flight network works

In the air the fingerprint says nothing, since a smooth cabin reads as standing still. Until the typical flight replaced it, a network read the speed from the phone's tilt (Section 3.1); a phone that turned over in a bag at the takeoff push read a tilt beyond anything it was trained on. The network now shipped is a correction on top of the typical flight: its eight inputs are the hours and the log of the minutes since the roll, whether the cabin has started down, the hours and log minutes since then, and the difference g the takeoff measured (the log of its speed over the typical one at the handover), as it is and fading over 5 and 20 minutes. Two layers of 32 ReLU units follow, and the single output is the log of the ratio to the typical speed. It was trained with an absolute-error loss, so it fits a median, on the same pooled flights. It is computed every second in the air and logged, not used (Section 4.4).

Table 6. A store still filling up against the built-in network, on motorcycle and car journeys, with each journey's own examples left out. *Where the app hands over from the network to the store.

	Speed error (km/h)			Distance vs GPS			Journeys
Speed model	All	Moto.	Car	All	Moto.	Car	within 20%
Store, 60 examples	10.2	10.2	10.2	−9.5%	−21.3%	+0.0%	56%
Store, 150 examples	10.1	9.9	10.4	−7.8%	−21.0%	+3.0%	57%
Store, 300 examples	9.5	9.5	9.4	−6.8%	−12.5%	−2.2%	65%
Store, 1,000 examples	8.9	9.0	8.7	−7.1%	−12.2%	−2.9%	72%
Store, 2,000 examples	8.6	8.9	8.2	−6.6%	−9.3%	−4.3%	72%
Store, 3,000 examples*	8.6	8.9	8.3	−4.7%	−7.7%	−2.2%	74%
Store, 4,000 examples	8.4	8.7	8.0	−7.6%	−11.5%	−4.4%	70%
Built-in network	8.1	8.8	7.5	−7.0%	−6.6%	−7.3%	79%

5.3 How well the speed network works

A new phone starts with the built-in network. On recordings it had never seen it missed by 8.8 km/h on a motorcycle and 7.5 km/h in a car, against 8.8 and 7.8 for the main model with a full store, and counted 93.4% and 92.7% of the distance against 90.2% and 94.1% (Figure 9). Band by band the two read speed alike: both squeeze the motorcycle's range, and both read a car close to GPS up to 60 km/h and low above it (Figure 9a–b). Journey by journey the network came closer to GPS on 38 journeys and the full store on 24, the rest within two points of each other (Figure 9c). A store still filling up does worse (Table 6, Figure 9d–f), above all on a motorcycle, whose rides a store of up to 1,000 examples reads an eighth to a fifth short: 10.2 km/h with 60 examples, 8.9 with 1,000 and 8.4 with 4,000, still a little behind the network's 8.1. The app hands over to the store at 3,000 examples all the same, because the store keeps adapting to the phone's own vehicle and pocket, an advantage these recordings, made with the same few vehicles the network was trained on, cannot show. Walking (99.2%) is nearly the same with either, since it is counted by steps; in the air neither is used.

6 Results across every kind of travel

Table 7 and Figure 10 set the four kinds of travel side by side. Direction is the stronger half everywhere: within 30° of GPS 79% of the time on a motorcycle, 83% in a car, 93% on foot and 72% of the aircraft's track on the plane. Distance is close in a car (−6.0%), on foot (−0.7%) and on the plane (95% and 94% of the two four-hour flights' tracks), and short on a motorcycle (−9.9%), where the speed model squeezes the range. Motorcycle and car together, the app added 619 km where GPS measured 671 km, 7.7% short.

Table 7. Every kind of travel side by side.

	Motorcycle	Walking	Car	Plane
Recorded	63 journeys	129 stretches	53 journeys	3 flights
GPS km checked	301	5.2	370	77 + 6,898 ADS-B
Distance against GPS	−9.9%	−0.7%	−6.0%	+1.3%; ADS-B −5, −6%
Speed	error 8.8 km/h	4.8 for GPS's 4.5 km/h	error 7.8 km/h	40/61/29, 80/44/102 km/h
Direction, median error	13°	10°	14°	13° (phones 8°)
Direction within 30°	79%	93%	83%	72%
Whole route, median	turned 9°	—	turned 14°	ends 335 km off

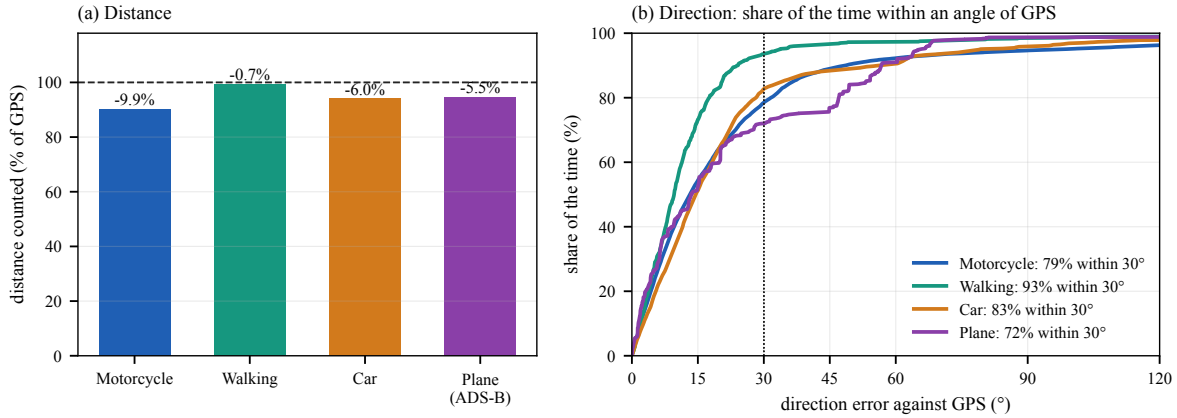


Figure 10. Every kind of travel side by side. (a) Distance counted, as a share of GPS's; for the plane, the two four-hour flights against their ADS-B tracks. (b) How often the direction is within a given angle of GPS, or of the aircraft's track over the six recordings on the three flights.

6.1 What each part of the direction method adds

Table 8 replays every recording with the direction method built up one part at a time, graded exactly as above. For cars the share of seconds within 30° goes from 63% with the heading and the angle from turns alone to 67% with drift measured at stops, 68% with right and left turns averaged apart, 68% with the push check, 71% with the clean magnetometer, 75% when drift is measured only in Earth's field and 83% when north is taken only from a field checked by turning; routes within 30° go from 73% to 86%, and routes of either vehicle turned more than 45° from 11 to 5. Drift measured at stops helps a drive where the phone sat in an Earth-like field and harms one where the field was disturbed (Section 7); counting only stops in Earth's field keeps the first and removes the second. For motorcycles every version is within 2 points of the others. The short flight, replayed with the speed the app has in the air, is within 30° 87% of the time with the first version and 99% once drift is measured at stops.

Table 8. What each part of the direction method adds: the share of graded seconds and of whole routes within 30°, routes turned more than 45°, and the flight's seconds within 30°. Each row adds one part to the row above. The magnetometer applies only where the magnetic field was recorded (20 car drives and 24 motorcycle rides so far).

	Motorcycle		Car		Routes	Flight
Direction method	Seconds	Routes	Seconds	Routes	over 45°	seconds
Heading + angle from turns	78%	86%	63%	73%	11	87%
+ drift at stops	77%	86%	67%	73%	10	99%
+ right, left apart	79%	88%	68%	80%	7	99%
+ turn explains push	79%	88%	68%	77%	7	98%
+ clean magnetometer	79%	88%	71%	80%	7	98%
+ drift only in Earth's field	79%	88%	75%	82%	7	98%
+ north checked by turning	79%	88%	83%	86%	5	98%

7 Discussion

7.1 Where it goes wrong

- **On a motorcycle, fast riding reads slow and slow riding reads fast.** Above 60 km/h the speed is under two-thirds of the true value (Table 2). Over a journey the two partly cancel, but a journey that is mostly fast comes out short.
- **A constant turn over a whole drive.** Much of the remaining direction error is one angle that lasts the whole drive: 35 of the 109 graded motorcycle and car recordings are turned by more than 15° on average. Removing one such turn per drive — which needs GPS, so it only bounds what is possible — would raise the share of seconds within 30° from 81% to 87% and cut the median error from 13.9° to 8.2°. It comes from where the heading started and from the angle learned for the phone, and the replay cannot separate the two. On the car drive with a clean magnetic field the magnetometer cut it from 34° to 22°, and the seconds within 30° rose from 16% to 86%.
- **The first minute of a ride.** Until the angle of the phone is learned the direction comes from the heading alone (Figures 2d and 4d). The redrawing fixes the saved route afterwards, but a very short ride may never learn the angle well.
- **A short, slow drive.** The angle is learned only while the speed model reads more than 14 km/h, so on a short drive in traffic it rests on a few turns, and which turns depends on the speed model. On one car drive, ten minutes at a median 9 km/h, the speeds the app read on the day kept 100% of the 87 graded seconds within 30° of GPS. Replayed with the speed model rebuilt from the other journeys, a different handful of turns sets the angle each time that model changes: in this re-run, the first with the present speed logic, 93% of those seconds were within 30°.
- **A phone that moves.** A hand shakes the phone and shaking reads as speed. A hand can no longer raise the speed, but a phone picked up while slowing keeps the speed it had when it was picked up, unless the trip is a pocket motorcycle; in the re-run the test for one called 58 of the 67 motorcycle journeys and 0 of the 54 car journeys a pocket motorcycle. A phone that shifts in the pocket turns the drawn route by about as much as it moved, until enough new turns have been learned.
- **A store full of another vehicle.** The store keeps 4,000 examples and each new one replaces a random old one, so a run of car drives leaves it mostly car. On the next two motorcycle rides it declined 44–47% of the riding seconds, and the app, which then held

its last answer, read 36% and 58% of GPS as recorded. The newest version answers those seconds with the starting network, which on the same seconds reads 64% and 88%. The re-run draws its store from every journey, so it cannot show the forgetting itself.

- **A disturbed field at stops.** On one 49-minute drive the field read about 80 μT at stops and 165 at speed. The stops then measured a drift that did not happen while moving, and the route turned further off every minute: with every still second counted, 44% of seconds were within 30°; with no drift at all, 88%; counting only stops in Earth's field, 92%. The bound matters: allowing up to 65 μT let the bad stops, at 62 to 65, back in. Only the recordings with the magnetometer logged, 45 so far, can show this at all, and the gain rests on that one drive.
- **A fast drive with nowhere to measure drift.** The drive home from an airport, 45 km and 55% of it above 80 km/h, stood still for only 20 seconds, and the field there and on the move was far from Earth's (42 to 278 μT). With no stop to measure the gyroscope's drift and no field to correct north, the heading turned by a steady 3.8° a minute for half an hour: 35% of its seconds were within 30°, the same live as in the re-run, and its fast stretches read low, so it counted 75% of the distance. Without it the cars' seconds within 30° would be 87%.
- **A field that looks like Earth's but is not.** On the same drive the steady-field test found seconds of Earth-like strength, steady for a minute, and took the correction from them; GPS shows it was about 30° wrong, and those seconds dipped 14–28° where Earth's field here dips about 35°. A steady field proves little while a car stands or runs straight. Checked sample by sample against GPS on the 45 recordings with the magnetometer logged, the minutes that passed the turn check put north within 15° of the truth 80% of the time (median error 8°, worst 68°, 1504 samples); the steady-field seconds did so 66% of the time (median 12°, worst 171°, 2280 samples). Over those recordings' graded seconds the turn check moved the share within 30° from 77% to 84%. Matching the field's strength and dip to Earth's values instead was fragile (Table 9).
- **A magnet beside the phone.** By a car's MagSafe charger the phone read up to 2,600 μT, fifty times Earth's field, yet reported its compass as well calibrated; the iPhone's own Compass then drifted 49° with the gyroscope. The app ignores such a field. On the drive with the charger beside the phone throughout, a few turning minutes still showed Earth's field, and with them the seconds within 30° rose from 0% to 93%; a phone that never leaves such a field cannot be corrected at all.
- **Car-park ramps and rides never recognised.** A car crawling up a concrete ramp shakes like one at 40 km/h, so distance is withheld while climbing and turning together; once that test fired on an ordinary road and dropped 9 seconds at 48 km/h, about 115 m. On 28 journeys the step counter took the engine for footsteps: as recorded, 14.2 km that GPS measured at 10 km/h or more were counted by steps as 2.0 km. On 7 of them that was a fifth of the ride or more, at worst 79%. Replayed with steps ending a ride only while the phone swings, 21 journeys still have such stretches (8.4 km by GPS). Those are left out of the comparisons (Section 3.5), so their loss is not in the distances reported above.
- **In the air, a typical speed.** After the takeoff the speed is what real airliners typically did at that point of a flight, not a measurement of this one. A headwind or tailwind, a faster or slower aircraft, a long hold or a short hop will make it read high or low: a twin-aisle jet reads several percent low in the cruise. Examples learned on the phone's own earlier flights with GPS take over when there are some, which three flights could not test. Turns made while iOS has paused the app are not seen, and a flight is not picked up again if iOS ends the app. A turn that iOS shortens on a phone lying still is made up

only where the magnetic field is steady, which it was not beside a charging tablet; and a phone handled through the takeoff starts from a roll it could not measure.

7.2 What did not work

Table 9 lists approaches that were built and measured against GPS on the same recordings and did no better. Two are worth a note. Leaving the speed out of the angle learner seemed attractive, since it would make direction independent of the speed model, but a car brakes into a corner and accelerates out of it, and a learner driven by the change in turn rate adds the two up instead of cancelling them. And weighting turns by the square root of their push drew individual drives very differently, better on one and worse on another, yet over the 66 rides then recorded the learned angle was the same (9.5° against 9.2° median error): single drives are a poor guide to a method.

7.3 Lessons

- **The phone's own definitions caused more error than the models.** Its built-in heading becomes unreliable in a common pocket position, drifts in a car and can start wrong; its "user acceleration" points the opposite way to the vehicle's acceleration. Each of these, once found, mattered more than any change to the estimators.
- **GPS can hide in decisions as well as in distance.** The no-GPS mode never drew a metre from GPS, yet at times it used GPS to decide between walking and riding, to set the first heading, or to carry an angle learned before the mode was switched on. Only reading every place the mode touched GPS found them.
- **Two errors can hide each other.** On one drive a worse angle estimate drew a better route, because it cancelled an error in the heading reference. Judge a part of the method against physics, not only by the route it draws.
- **A replay must do everything the app does,** stops and a phone in the hand included; with those rules in, it differs from what the app drew by a median 5%.
- **A replay must feed each part what the app feeds it.** The flight's direction was at first replayed with the cabin's near-zero vibration speed rather than the speed the app has in the air. That starved the angle learner, and the result in the air swung between 4° and 24° with nothing but the replay's random choices; with the app's own speed it holds at 10–13°.
- **A replay's own randomness can move results.** When a recording was added, drawing every journey's store afresh moved numbers that had nothing to do with it by several points. Each example now keeps its own fixed chance of being drawn, so a new recording changes each store only by its own examples — which, small as they are, still move single journeys (Section 8).

8 Limitations

All ground recordings come from one person and one iPhone, mostly in one city; the three flights were recorded on that phone and, on the two long ones, on a second phone and a tablet. Other people, phones and vehicles may behave differently, the starting network most of all. The typical flight is learned from other aircraft on other days and checked on only three flights, two of them out and back on one route in one aircraft type. Most results replay the newest method on earlier recordings rather than coming from live use, with a speed store drawn from

Table 9. Approaches that were measured and did not do better, each on the recordings available when it was tried.

Approach	Idea	Measured against GPS
Direction		
Push when speeding up or braking	In a straight line the push points forward	44° off
Axis the motorcycle leans about	The lean axis follows the direction of travel	25° off
Direction of the rider's thigh	The thigh points forward while seated	73° off
Angle from turns without the speed	Weight each second by the change in turn rate	15–40° off on one drive
Angle from full physics	Add the speed model's change in speed to the turn	cars 80% → 48% within 30°
Square-root turn weights	Let the many small bends count more	a tie over 66 rides
Median instead of mean for drift	Ignore short bursts at stops	walking 93% → 88%
Looser stillness for drift	Measure drift at more stops	motorcycles 73% → 68%
Magnetic disturbance subtracted	Earth's field turns with the car, a magnet fixed to the phone does not; fitted with a gyroscope-only heading	leftover 40–200 μT: the field changed even on straight roads
Field matched to Earth's numbers	A second is clean if its strength and dip match Earth's here	right only ∼60% of the time; 3 μT or 3° off moved one drive 96% → 68%
Core Motion's pulls undone	In a disturbed field, turn back whatever north the magnetometer adds	no change: after its first seconds it pulls under 5° a minute
Walking, riding and the air		
Thigh swing for walking direction	Swing is sharper forwards than backwards	wrong more often than right
Other riding-or-walking tests	A steady push; speed overruling steps	blocked 5% of real walking
Speed vetoing bump steps	The model's or the recent riding speed overrules steps on a motorcycle	blocked real walks: the held speed is still high after getting off
Fading the held speed in a car too	A phone taken in hand means a stop	cars in the hand +21% → −37% of GPS: a passenger holds it at speed
Speed from turns in the air	Sideways push over turn rate	phone's turning hid the aircraft's
Integrating the whole climb	Carry the takeoff integration on	worked for one setting only
Cabin shaking as a fifth input	Shaking changes with height and speed	68% → 69% of NASA flights within 10%
Speed		
Last ten seconds into the network	More context for each answer	better per second, not per journey
Traffic around slow riding	Share of the last minutes stopped, recent speed	no signal: 82–112% of GPS in every band
Magnetic flicker for crawling	The field flickers more at a crawl than at idle	idle and crawl told apart, but fixed at most 340 m on 8 rides and shrank fast riding

the same person's other journeys; on a short drive the replayed and the live speeds can teach different angles (Section 7.1). Single journeys depend on the store's exact contents: adding one eight-minute car drive changed about 1% of each journey's store and moved single journeys' distance by a median 3.6 points (up to 14), and the motorcycle and car totals by 0.3 and 1.3 points, so the shares of journeys within 20% move by several journeys from one re-run to the next. GPS in a pocket is accurate only to tens of metres, so small errors cannot be measured reliably. In every journey above the phone rode in a pocket, lay flat or sat on a mount. On one car drive it was held in the hand the whole way; recorded live with the current version and kept out of every result, that drive counted 77% of the distance GPS measured and missed the speed by 13.7 km/h, and above 50 km/h it read 31 km/h where GPS said 65. The rule for a phone in the hand applies only while it is being moved: the stretches it marked read 83% of GPS, and those held steadily, which it did not mark, 66%. The magnetic field was recorded on 20 car drives and 24 motorcycle rides and one other recording so far, and everything said here about it rests on those 45. On the first drive a MagSafe charger sat near the phone for part of the time and the correction worked: 83% of seconds within 30° with the steady-field test, 94% with the turn check. On the second it sat there throughout; the field was steady at an Earth-like strength for only 51 seconds, short of the minute the steady-field test needs, while turning minutes passed the turn check at 27 seconds (0% → 93%). On the third, with the phone placed away from the charger, the field swung between 32 and 74 μT from minute to minute and was never steady; the turn check passed at 7 seconds and the share within 30° moved from 79% to 76%. On the fourth the field rose with speed, from about 70 to 170 μT, and changed even on straight roads, which is why subtracting it as a fixed disturbance did not work (Table 9); no minute passed the turn check, so the steady-field test decides that drive (92%). On the fifth, 52 minutes long, the turn check passed at 169 seconds, most of them in the first minutes, and put north close to where GPS says it was; but once those samples were twenty minutes old and dropped, a few scattered later ones moved the correction the wrong way for the rest of the drive. The share within 30° stayed high (99%; 100% with no magnetometer at all), because on that drive the angle learned from turns carried an error of the opposite sign. On the sixth the turn check passed at 152 seconds, most of them in the first minutes, and its correction agreed with GPS to within a few degrees, though the drive's share within 30° was 85% against 97% with no magnetometer at all. On the first motorcycle ride the field was steady for 88 seconds and the correction took the graded seconds within 30° from 77% to 78% (73% once the turn check, passing at 5 seconds, takes over), though weak GPS left only 74 seconds to grade; on the second the field stayed near 500 μT, ten times Earth's, nothing could correct the heading, and 24% of its seconds were within 30°. On the third it swung between about 45 and 150 μT while riding and passed the turn check only on the walk afterwards, so the ride's direction rested on its turns alone (83% within 30°). On the fourth it ran from about 40 to over 1,000 μT, was steady for only 13 seconds and never passed the turn check, and its direction too came from its turns (90%).

9 Conclusion

A phone's own sensors can draw a usable route without satellites. Direction is the stronger half: learned from the vehicle's turns on a heading kept honest by its stops and, when possible, by the magnetometer, it stays within 30° of GPS most of the time on every kind of travel measured. Speed from vibration is close in a car and on foot but compressed on a motorcycle. In the air, the takeoff measured by the phone and then the typical speed of public airline flight data counted two four-hour flights to within 5% and 6% of their ADS-B tracks and a short one to within GPS's distance, and its direction, from the runway and the gyroscope checked against the magnetic field, was typically 13° off. The largest remaining error is a constant turn over a

whole drive, from where the heading started and from the angle learned for the phone, which so far only a clean magnetic field has been able to reduce.

Code and data. The app, including the method described here, is open source at https://github.com/yu314-coder/GPS-location-app. The recordings contain the author's locations and are not published; the airline flight data is public [9, 10], and the typical flight and both built-in networks ship with the app.

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A Every journey

Each motorcycle (M) and car (C) journey, numbered in the order recorded, with no dates. Minutes and GPS km are what GPS could check; app/GPS is the distance ratio for journeys of at least 1 km; speed error is the average over half-minutes; direction is graded where at least 50 seconds could be; the route columns are the whole-route comparison. A dash means too little GPS to grade.

Table 10. Every motorcycle (M) and car (C) journey, in the order recorded.

	Min.	GPS km	App/GPS (%)	Speed err. (km/h)	Dir. within 30° (%)	Dir. median (°)	Route km	Turned (°)	Size
M01	20	5.8	66	12.8	83	10	13.0	-1	0.63
M02	12	5.0	41	24.0	87	14	6.6	+12	0.46
M03	20	3.5	65	10.9	49	32	9.9	-15	0.47

	Min.	GPS km	App/GPS (%)	Speed err. (km/h)	Dir. within 30° (%)	Dir. median (°)	Route km	Turned (°)	Size
M04	17	1.4	33	6.8	63	11	5.1	+6	0.76
M05	16	2.5	128	5.5	84	16	4.8	-7	1.04
M06	19	4.1	104	8.0	48	32	12.3	+10	0.79
M07	20	5.8	79	11.2	85	15	8.5	+4	0.76
M08	24	7.4	72	10.3	60	26	13.8	-26	0.98
M09	9	3.1	112	4.2	91	9	3.9	-12	1.07
M10	27	9.2	86	9.8	89	13	14.7	-15	0.90
M11	20	5.0	82	7.9	76	19	11.9	+10	0.73
M12	14	4.8	92	6.5	73	24	7.3	+19	0.80
M13	23	6.8	92	6.5	56	25	11.9	+27	0.98
M14	18	5.7	93	9.2	84	10	9.5	-8	0.85
M15	18	8.2	84	12.3	90	6	12.0	-2	0.98
M16	16	6.1	93	8.8	82	18	8.0	-23	0.96
M17	–	–	–	–	–	–	2.5	-51	0.58
M18	4	1.4	101	4.8	100	3	2.0	-2	0.87
M19	14	6.1	77	8.5	40	32	7.3	+32	0.74
M20	19	6.2	97	8.5	90	5	9.3	-11	0.76
M21	15	4.0	106	8.1	78	15	8.8	+7	0.78
M22	10	2.4	109	7.2	85	8	5.7	+5	0.81
M23	11	3.1	88	9.3	83	7	6.5	-9	0.92
M24	20	3.5	88	9.7	87	7	12.0	-5	0.94
M25	9	3.5	100	6.3	97	4	4.1	-1	1.11
M26	27	9.7	90	9.3	77	10	16.6	-2	0.88
M27	–	–	–	–	–	–	2.7	-102	0.39
M28	14	2.4	129	8.7	61	18	8.0	-3	1.93
M29	8	2.8	82	9.0	92	7	4.0	-11	0.75
M30	9	0.6	–	0.9	34	42	3.2	-41	0.69
M31	25	8.3	68	11.0	66	15	15.6	-13	0.75
M32	11	3.6	94	7.7	77	18	4.3	-16	0.83
M33	–	–	–	–	86	13	–	–	–
M34	15	1.7	96	4.4	97	7	5.1	-7	0.59
M35	32	6.8	97	6.2	23	33	11.7	-37	0.91
M36	12	2.6	98	6.1	65	18	6.0	-9	1.17
M37	19	6.3	99	10.9	87	10	13.9	+9	0.79
M38	21	4.1	95	4.5	75	17	11.5	-22	0.79
M39	8	3.1	94	8.2	79	24	5.7	-13	1.06
M40	–	–	–	–	20	53	1.7	+61	1.02
M41	27	5.7	79	9.6	58	21	13.1	-4	0.86
M42	20	4.9	98	12.9	79	19	10.4	-18	1.06
M43	20	5.6	92	9.3	80	8	10.3	-4	0.95
M44	7	1.6	110	5.4	73	18	3.1	-26	0.92
M45	7	1.8	113	10.5	24	50	3.3	+32	1.27
M46	19	7.6	78	11.0	83	8	11.0	-6	0.84
M47	17	5.5	99	7.5	90	4	7.5	-3	0.85
M48	28	5.1	110	6.6	66	28	13.3	-21	1.15
M49	8	1.8	52	18.6	60	9	5.0	-173	1.61
M50	20	5.5	102	7.5	89	9	7.9	+6	0.89
M51	18	5.8	96	4.4	99	4	7.4	-2	0.88
M52	20	5.4	106	7.1	78	23	8.5	+15	0.73
M53	12	3.4	112	7.9	83	13	3.8	-21	1.04
M54	9	2.3	96	9.5	99	8	3.1	+7	0.85
M55	19	5.9	97	6.8	93	9	8.5	+1	0.85
M56	13	5.7	96	10.2	94	6	7.8	+1	0.85
M57	17	6.6	97	7.8	97	4	8.3	+3	0.91
M58	12	2.7	81	13.7	81	9	7.0	+8	0.76

	Min.	GPS km	App/GPS (%)	Speed err. (km/h)	Dir. within 30° (%)	Dir. median (°)	Route km	Turned (°)	Size
M59	15	4.8	75	10.1	35	38	11.0	+22	0.75
M60	15	3.7	96	5.2	75	19	7.0	-3	1.02
M61	31	9.3	92	8.5	95	11	13.8	+9	0.85
M62	14	5.2	71	15.1	84	15	7.4	+7	0.83
M63	18	5.5	78	6.9	91	15	6.5	+10	0.72
M64	22	5.2	121	12.6	73	17	12.0	+6	1.07
M65	17	5.7	89	7.1	97	11	7.4	+11	0.86
M66	18	6.2	84	9.5	83	5	10.1	+2	0.87
M67	18	6.5	104	8.0	95	6	8.1	-4	1.02
C01	11	6.6	117	11.3	–	–	–	–	–
C02	16	7.6	138	13.7	–	–	–	–	–
C03	14	8.1	138	21.2	–	–	–	–	–
C04	9	2.7	90	8.0	–	–	–	–	–
C05	13	6.5	57	17.7	–	–	–	–	–
C06	24	7.5	90	8.4	–	–	–	–	–
C07	16	6.5	85	9.9	–	–	–	–	–
C08	29	6.0	117	8.9	–	–	–	–	–
C09	25	9.5	79	7.8	65	17	13.7	+28	0.66
C10	24	7.4	64	10.0	89	15	10.9	+16	0.98
C11	3	0.8	–	2.9	100	5	1.6	+9	1.12
C12	21	6.5	119	6.7	97	5	9.0	-3	1.14
C13	12	3.5	129	6.6	98	6	4.4	-7	1.27
C14	10	2.5	98	2.4	100	18	3.2	+26	0.91
C15	12	2.6	107	3.1	62	27	3.1	-28	0.94
C16	10	1.3	119	6.3	49	33	2.1	-10	0.82
C17	6	1.1	98	10.0	37	35	1.6	+26	1.16
C18	7	0.9	–	5.1	–	–	2.0	+39	1.32
C19	8	1.1	140	5.0	87	25	2.7	+30	0.93
C20	19	7.8	107	6.6	94	14	9.5	-14	0.99
C21	6	1.2	129	6.6	89	13	2.5	-1	0.93
C22	15	9.1	94	5.3	98	19	10.4	-13	1.02
C23	25	13.6	100	5.5	84	26	16.3	-24	0.99
C24	8	1.1	157	8.5	96	13	1.8	+25	1.10
C25	4	0.9	–	7.8	76	9	1.2	+39	1.11
C26	15	3.0	92	5.1	67	26	4.7	+28	1.03
C27	12	4.3	120	6.1	99	8	4.9	+11	1.17
C28	11	1.9	117	4.1	99	10	2.9	-9	1.16
C29	18	5.5	104	2.9	1	62	6.5	+61	1.03
C30	10	2.5	103	2.6	84	24	2.8	-25	1.00
C31	15	2.2	100	2.6	96	10	2.8	+10	1.00
C32	11	0.0	–	1.2	–	–	–	–	–
C33	13	2.0	110	2.9	100	6	2.7	-4	1.06
C34	32	9.2	98	3.7	99	9	11.0	-8	0.93
C35	34	10.0	97	2.6	94	16	11.2	-17	0.94
C36	10	1.3	109	2.7	93	19	1.7	-16	1.01
C37	8	1.7	103	3.3	76	15	2.1	-11	0.86
C38	49	21.9	104	5.7	92	22	24.5	-14	1.20
C39	52	22.9	95	4.8	99	8	25.4	+2	0.81
C40	22	6.8	92	6.2	85	4	7.8	-1	0.90
C41	35	6.9	91	3.4	95	2	7.9	-1	0.85
C42	18	5.9	111	5.1	83	13	6.7	-23	1.18
C43	7	0.7	–	11.2	100	3	1.4	-4	0.79
C44	34	7.6	91	3.2	100	14	8.9	+14	0.89
C45	26	14.5	98	3.9	97	15	15.7	-13	0.90
C46	11	3.4	115	4.4	91	12	4.0	+11	0.96

	Min.	GPS km	App/GPS (%)	Speed err. (km/h)	Dir. within 30° (%)	Dir. median (°)	Route km	Turned (°)	Size
C47	37	12.3	100	5.2	78	15	15.3	-23	0.95
C48	4	0.7	–	5.0	99	13	1.2	-8	1.09
C49	71	41.8	65	19.8	80	11	50.6	-30	0.47
C50	–	–	–	–	80	15	–	–	–
C51	16	4.5	97	3.4	99	4	4.9	+1	0.95
C52	15	4.9	102	3.1	98	5	5.7	+5	1.02
C53	14	4.6	101	3.2	100	9	5.0	-5	1.01
C54	37	44.7	75	22.7	35	49	46.5	+41	0.51

B Settings

Table 11. Every setting of the method as evaluated.

Setting	Value
Speed	
Vibration window for one fingerprint	5 s at 50 Hz, 11 numbers
Store of the phone's own examples	up to 4,000
Neighbours averaged	12
Built-in network	$11 \to 32 \to 32 \to 1$; used until the store holds 3,000, and after that whenever the store declines
Network refuses a fingerprint	farther from all training clusters than 99.5% of training data
A phone in the hand	smoothed rotation (3 s) above 0.40 rad/s: the speed may fall, not rise, and is held
Held speed fades	on a pocket motorcycle, $\min(v, v_0 e^{-t/10\,\mathrm{s}})$; a pocket motorcycle when the unhandled seconds above 20 km/h (at least 60, the last 900) turn at a median ≥ 0.09 rad/s and the phone is not upright (pitch $> 0°$ with $\lvert\text{roll}\rvert < 30°$)
Takeoff integration in the air	from the last still moment, up to 120 s
Takeoff test	30 kt, then 200 km/h within 30 s
Typical flight	median of 1,496 flights each minute since the roll, then 831 km/h; by minute since the descent
Takeoff carried	86, 58, 29, 32, 15, 6, 2% at $\frac{1}{2}$, 1, 2, 3–4, 6, 10, 20 min
Cabin as a clock	climb 400 m in 10 min; descent 300 m below the peak; landed within 60 m or 3%, not falling for 30 s
Direction in the air	roll push over 35 s, a runway's within 40°; turns felt (load over 1.2%) or steady (0.12°/s for 15 s); unfelt over 2 min is drift; field steady within 30%
Walking: a step's echo refused	while the last 4 s show a clear rhythm (autocorrelation ≥ 0.3 at 0.3–0.8 s), a peak inside 0.65 of its step period
Steps end a ride	while a vehicle is current, only with the smoothed rotation at 0.60 rad/s or more
Direction	
Riding begins	9 km/h (2.5 m/s)
Learning the phone's angle	above 14 km/h (4 m/s); turn explains $\geq 10\%$ of the push
First estimate	after 30 s and a summed speed × turn rate of 20 m/s^2
Redrawing a ride	when the angle moves $\geq 3°$, at most every 10 s
A new ride	after 20 s without riding
A phone taken out of the pocket	angle forgotten after 10 s in the hand
Drift measured at stops	stopped, turning < 0.005 rad/s, hand-rotation < 0.2 rad/s
Drift estimate	after 30 still seconds, 10-minute memory, at most 3°/min
Drift counted only in a field of	30–60 μT
Clean magnetic field	25–65 μT, strength and dip steady to 3 over 60 s
North checked by turning	last 60 s: heading swept $\geq 30°$, field in the world frame within 8 μT (RMS), 25–70 μT
Magnetometer correction	any turn-checked minute in the last 20 min; before the first, 60 clean seconds; spread $< 15°$
Walking direction window	4 s